Linear Algebra and Its Applications

Section 4.3: Linearly Independent Sets; Bases

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

4.3 Linearly independent sets; bases 3

Practice problem 4.3.P1: A basis must have the right ambient space 3

Practice problem 4.3.P2: Remove redundant spanning vectors 3

Practice problem 4.3.P3: Spanning every vector of a line is not enough 4

Exercise 4.3.1: Identify a basis using original pivot columns 4

Exercise 4.3.2: Identify a basis using original pivot columns 4

Exercise 4.3.3: Identify a basis using original pivot columns 5

Exercise 4.3.4: Identify a basis using original pivot columns 5

Exercise 4.3.5: Identify a basis using original pivot columns 6

Exercise 4.3.6: Identify a basis using original pivot columns 6

Exercise 4.3.7: Identify a basis using original pivot columns 7

Exercise 4.3.8: Identify a basis using original pivot columns 7

Exercise 4.3.9: A null-space basis from one free variable 8
Exercise 4.3.10: A null-space basis in five coordinates 8
Exercise 4.3.11: A basis for a plane 9
Exercise 4.3.12: A basis for a line 9
Exercise 4.3.13: Null space and original column space 9
Exercise 4.3.14: Two different spaces from one reduction 10
Exercise 4.3.15: Identify a basis using original pivot columns 11
Exercise 4.3.16: Identify a basis using original pivot columns 11
Exercise 4.3.17: Identify a basis using original pivot columns 12
Exercise 4.3.18: Identify a basis using original pivot columns 12
Exercise 4.3.19: Delete a generator using a known relation 13
Exercise 4.3.20: Another known dependence relation 13
Exercise 4.3.21: Basis definitions and row operations 13
Exercise 4.3.22: Minimal spanning and maximal independence 14
Exercise 4.3.23: Four spanning vectors in four dimensions 14
Exercise 4.3.24: An independent square family 14
Exercise 4.3.25: A proposed basis outside the subspace 14
Exercise 4.3.26: A trigonometric spanning set 15
Exercise 4.3.27: The mass–spring vibration space 15
Exercise 4.3.28: A critically damped circuit space 15
Exercise 4.3.29: Too few vectors to span 15
Exercise 4.3.30: Too many vectors to be independent 15
Exercise 4.3.31: Dependence survives a linear map 15
Exercise 4.3.32: Injectivity reflects dependence 16
Exercise 4.3.33: Independence of two quadratic polynomials 16
Exercise 4.3.34: A polynomial dependence by inspection 16
Exercise 4.3.35: Construct a four-vector family with a two-vector basis 16
Exercise 4.3.36: Bases of two spaces and their sum 17
Exercise 4.3.37: Four independent functions by exact evaluations 18
Exercise 4.3.38: Seven cosine powers are independent 19

4.3 Linearly independent sets; bases

A basis for H must consist of vectors *in* H, span exactly H, and be linearly independent. Spanning a larger space is not enough. Independence means that the only zero combination has every coefficient zero. Removing a generator already in the span of the others does not change the span.

For a matrix, row operations preserve the solutions of its homogeneous system and all linear relations between columns. Consequently the pivot indices in RREF identify which *original* columns form a basis for the column space. Reduced columns themselves need not belong to the original column space. In the numerical work below, columns are denoted v_j in their printed order; a displayed RREF and the explicit column relations provide checkable elimination and spanning certificates.

Practice problem 4.3.P1: A basis must have the right ambient space

The two vectors are not scalar multiples: the first would require multiplier -2, but then its second entry would be 4, not 7. They are independent, but two vectors cannot span $\mathbb{R}^3$ because a 3×2 coefficient matrix cannot have a pivot in every row. Nor can they be a basis of $\mathbb{R}^2$: they have three coordinates and are not elements of that space.

Conclusion. They are a basis for their two-dimensional span in $\mathbb{R}^3$, not for $\mathbb{R}^3$ or $\mathbb{R}^2$.

Practice problem 4.3.P2: Remove redundant spanning vectors

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 6 & 2 & -4 \\ -3 & 2 & -2 & -8 \\ 4 & -1 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{4}{5} & 2 \\ 0 & 1 & \frac{1}{5} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (\frac{4}{5})v_1 + (\frac{1}{5})v_2$$
$$v_4 = (2)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Practice problem 4.3.P3: Spanning every vector of a line is not enough

Neither $e_1 = (1,0,0)^T$ nor $e_2 = (0,1,0)^T$ lies in $H = \{(s,s,0)^T : s \in \mathbb{R}\}$, since their first two coordinates differ. Their span is the entire xy-plane, which is larger than H. Thus they cannot be a basis of H, despite the valid expression $(s,s,0)^T = se_1 + se_2$. A basis of H is instead $\{(1,1,0)^T\}$: it is nonzero and its multiples are exactly H.

Conclusion. No. Membership in H and equality of the span with H are missing.

Exercise 4.3.1: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.2: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_2 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.3: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1 \\ -3 & -4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (1)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.4: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 2 & -8 \\ -1 & -3 & 5 \\ 1 & 2 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.5: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 3 & -3 & 0 & 0 \\ -3 & 7 & 0 & -3 \\ 0 & 0 & 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.6: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ -4 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 6 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.7: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 6 \\ 3 & -1 \\ 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -1 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.8: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ -2 & 3 & -1 & 0 \\ 3 & -1 & 5 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.9: A null-space basis from one free variable

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & 1 & 4 \\ 3 & -1 & -7 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The free variable columns are 3. The pivot equations are

$$\begin{aligned} x_1 &= 2x_3 \\ x_2 &= -x_3 \\ x_4 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.10: A null-space basis in five coordinates

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 1 & -2 & 1 & 5 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -8 & 0 & 16 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & -2 \end{bmatrix}.$$

The free variable columns are 4, 5. The pivot equations are

$$\begin{aligned} x_1 &= -2x_4 - 3x_5 \\ x_2 &= x_4 + 2x_5 \\ x_3 &= 2x_5 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.11: A basis for a plane

Solve the single homogeneous equation for x: $x = 3y - 2z$. Then

$$(x, y, z)^T = y(3, 1, 0)^T + z(-2, 0, 1)^T.$$

Every vector in the plane has this form. The displayed vectors lie in the plane, and a zero combination has second coordinate $y = 0$ and third coordinate $z = 0$. Thus they are independent and span precisely the plane.

Conclusion. $\{(3, 1, 0)^T, (-2, 0, 1)^T\}$ is a basis.

Exercise 4.3.12: A basis for a line

The line equation $y = -3x$ means that every vector is $(x, -3x)^T = x(1, -3)^T$. The vector $(1, -3)^T$ is nonzero, so the one-element set containing it is independent. Its multiples are exactly the line, so it spans the required space.

Conclusion. $\{(1, -3)^T\}$ is a basis.

Exercise 4.3.13: Null space and original column space

Null space: use the row-equivalent matrix.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & \frac{5}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 3, 4. The pivot equations are

$$x_1 = -6x_3 - 5x_4$$
$$x_2 = -\frac{5x_3}{2} - \frac{3x_4}{2}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -6 \\ -\frac{5}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: return to the original matrix.

The pivot indices are 1 and 2, so a column-space basis is $\{(-2, 2, -3)^T, (4, -6, 8)^T\}$. Do not take columns of the echelon matrix instead: row operations need not preserve the actual column space.

Exercise 4.3.14: Two different spaces from one reduction

Null space: solve the homogeneous equations.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 2 & 3 & -4 & 8 \\ 1 & 2 & 0 & 2 & 8 \\ 2 & 4 & -3 & 10 & 9 \\ 3 & 6 & 0 & 6 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 2, 4. The pivot equations are

$$\begin{aligned} x_1 &= -2x_2 - 2x_4 \\ x_3 &= 2x_4 \\ x_5 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: retain the original pivot columns.

The pivot indices are 1,3,5. Thus a basis for $\operatorname{Col} A$ is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 8 \\ 8 \\ 9 \\ 9 \end{bmatrix} \right\}.$$

Exercise 4.3.15: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 2 & 3 \\ 0 & 1 & -2 & -1 & -1 \\ -2 & 2 & -8 & 10 & -6 \\ 3 & 3 & 0 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 10 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -6 \\ 9 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (-2)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.16: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 & 3 & 5 & 2 \\ 0 & 0 & -1 & -3 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 1 & 2 & -1 & -4 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -\frac{5}{2} & 0 \\ 0 & 1 & 0 & \frac{3}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-\frac{5}{2})v_1 + (\frac{3}{4})v_2 + (3)v_3$$
$$v_5 = (\frac{1}{2})v_2 + (1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Exercise 4.3.17: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 4 & -2 & 8 & -8 \\ 0 & 0 & -4 & 4 & 4 \\ -4 & 2 & 0 & 8 & 0 \\ -6 & -4 & 1 & -3 & 0 \\ 0 & 4 & -7 & 15 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 0 \\ -4 \\ -6 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 2 \\ -4 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 0 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -8 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-1)v_1 + (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.18: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -3 & 3 & 0 & 6 & -6 \\ 2 & 0 & 2 & -2 & 3 \\ 6 & -9 & -4 & -14 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ -7 & 6 & -1 & 13 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -3 \\ 2 \\ 6 \\ 0 \\ -7 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -9 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -4 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -6 \\ 3 \\ 0 \\ -1 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.19: Delete a generator using a known relation

The supplied relation gives $v_3 = (4/3)v_1 + (5/3)v_2$. Therefore every combination of all three vectors is a combination of v_1, v_2, so those two still span H. They are not scalar multiples: if $v_2 = cv_1$, the first coordinate gives $c = 1/4$, but the second would be $-3/4$ rather than 9. Thus they are independent.

Conclusion. $\{(4, -3, 7)^T, (1, 9, -2)^T\}$ is a basis for H.

Exercise 4.3.20: Another known dependence relation

Rearrange $2v_1 - v_2 - v_3 = 0$ as $v_3 = 2v_1 - v_2$. The third generator is redundant, so $H = \text{Span}\{v_1, v_2\}$. If $v_2 = cv_1$, the first coordinate gives $c = 4/3$, while the second coordinate would require $3 = (4/3)4 = 16/3$, a contradiction. Hence the two retained vectors are independent.

Conclusion. $\{(3, 4, -2, -5)^T, (4, 3, 2, 4)^T\}$ is a basis for H.

Exercise 4.3.21: Basis definitions and row operations

(a) False. A singleton $\{v\}$ is independent whenever $v \neq 0$, because $cv = 0$ then forces $c = 0$; only the zero singleton is dependent.

(b) False. Spanning is only one requirement. The set $\{e_1, 2e_1\}$ spans a line but is dependent and hence is not a basis.

(c) True. An invertible square matrix has a pivot in every row and column, so its columns both span $\mathbb{R}^n$ and are independent.

(d) False. One can append redundant vectors to a spanning set. A basis is a *minimal* spanning set, equivalently a maximal independent set, not a largest spanning set.

(e) False. If A is replaced by an elementary row-equivalent matrix EA, the relations $Ac = 0$ and $EAc = 0$ are equivalent because E is invertible. Thus precisely the same coefficient relations remain.

Exercise 4.3.22: Minimal spanning and maximal independence

(a) False. $\{e_1\}$ is independent in $\mathbb{R}^2$ but does not span it.

(b) True. Repeatedly delete a vector expressible in terms of the others. Each deletion preserves the span, and finiteness ensures the process stops at an independent spanning subset, hence a basis.

(c) True, with "as large as possible" meaning that no further vector of the space can be added while preserving independence. If an independent set did not span, a vector outside its span could be adjoined without creating dependence. Thus a maximal independent set spans and is a basis.

(d) False. The usual free-variable construction always gives independent vectors: in their free-coordinate positions they have distinct unit entries.

(e) False. Use pivot columns of A, not of its row-reduced matrix. For example $A = (1, 1)^T$ reduces to $B = (1, 0)^T$; their column spaces are different lines.

Exercise 4.3.23: Four spanning vectors in four dimensions

Put the four spanning vectors into a 4×4 matrix A. Because they span $\mathbb{R}^4$, its echelon form has a pivot in every row. There are four rows, hence four pivots; all four columns are pivot columns. Therefore $Ac = 0$ has no free variable and only the zero solution. The columns are independent as well as spanning.

Conclusion. The four vectors form a basis for $\mathbb{R}^4$.

Exercise 4.3.24: An independent square family

The n independent columns of their $n \times n$ matrix give a pivot in every column. Hence there are n pivots, one in each row too. A pivot in every row means the columns span $\mathbb{R}^n$. They were already independent, so they satisfy both basis requirements.

Conclusion. $\{v_1, \ldots, v_n\}$ is a basis for $\mathbb{R}^n$.

Exercise 4.3.25: A proposed basis outside the subspace

A basis of H must be a subset of H. Here $v_1 = (1, 0, 1)^T$ and $v_3 = (0, 1, 0)^T$ have unequal second and third coordinates, so neither belongs to H. The supplied expansion does indeed express every $(s, t, t)^T$ using the three vectors, but it does not show that their span is contained in H. Their span is larger: the three vectors are actually independent and span $\mathbb{R}^3$. A correct basis of H is $\{(1, 0, 0)^T, (0, 1, 1)^T\}$, since $(s, t, t)^T = s(1, 0, 0)^T + t(0, 1, 1)^T$ uniquely.

Conclusion. No. Unique representation of the vectors of H does not repair the failure of the proposed basis vectors to lie in H.

Exercise 4.3.26: A trigonometric spanning set

Use the identity $\sin 2t = 2\sin t\cos t$. It shows the third function is half the second, so $\sin t$ and $\sin 2t$ span the same subspace as all three. If $a\sin t + b\sin 2t = 0$ for every t, evaluate at $t = \pi/2$ to obtain $a = 0$, then at $t = \pi/4$ to obtain $b = 0$. Thus the two retained functions are independent.

Conclusion. $\{\sin t, \sin 2t\}$ is a basis of the specified function space.

Exercise 4.3.27: The mass–spring vibration space

Exercise 4.1.19 gives all displacements as $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$, with fixed physical frequency $\omega > 0$. This already proves spanning by the two displayed functions. If $a\cos(\omega t) + b\sin(\omega t) = 0$ identically, $t = 0$ gives $a = 0$, and $t = \pi/(2\omega)$ gives $b = 0$.

Conclusion. $\{\cos(\omega t), \sin(\omega t)\}$ is a basis for the vibration space. If a nonphysical zero frequency were allowed, the sine function would vanish and the space would instead have basis $\{1\}$.

Exercise 4.3.28: A critically damped circuit space

The given description $v(t) = e^{-bt}(c_1 + c_2 t)$ means $H = \text{Span}\{e^{-bt}, te^{-bt}\}$. To check independence, suppose $ae^{-bt} + cte^{-bt} = 0$ for every t. Since $e^{-bt} > 0$, division gives $a + ct = 0$ identically. At $t = 0$ we get $a = 0$; at $t = 1$ we then get $c = 0$.

Conclusion. $\{e^{-bt}, te^{-bt}\}$ is a basis for the critically damped circuit's solution space H.

Exercise 4.3.29: Too few vectors to span

Place the $k < n$ vectors into an $n \times k$ matrix. At most k pivots are possible, so at least one of the n rows has no pivot. By the spanning criterion of Section 1.4, the columns cannot span $\mathbb{R}^n$. A basis must span the whole space.

Conclusion. A set with fewer than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.30: Too many vectors to be independent

There are $k > n$ columns but only n rows. The homogeneous system with these columns therefore has at least one free variable, so it has a nonzero coefficient solution. This is a nontrivial dependence relation among the k vectors. A basis must be independent.

Conclusion. A set with more than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.31: Dependence survives a linear map

Dependence provides scalars $c_1, \ldots, c_p$, not all zero, such that $\sum_j c_j v_j = 0$. Apply the linear map T:

$$\sum_j c_j T(v_j) = T\left(\sum_j c_j v_j\right) = T(0) = 0.$$

The coefficients have not changed, so they are still not all zero. This is a nontrivial dependence relation for the image vectors. Taking the contrapositive, independent images force the original vectors to be independent.

Conclusion. Linear maps preserve dependence, although they can turn an independent family into a dependent one.

Exercise 4.3.32: Injectivity reflects dependence

A dependence relation among the images has the form $\sum_j c_j T(v_j) = 0$ with not all c_j zero. By linearity,

$$T\left(\sum_j c_j v_j\right) = 0 = T(0).$$

Injectivity says equal images have equal inputs, so $\sum_j c_j v_j = 0$. The same nonzero coefficient list proves original dependence. The contrapositive states that an injective linear map carries an independent family to an independent image family.

Conclusion. Injectivity prevents a nonzero input combination from being collapsed to zero.

Exercise 4.3.33: Independence of two quadratic polynomials

Suppose $a(1+t^2) + b(1-t^2) = 0$ as a polynomial identity. Its constant coefficient is $a+b$, and its t^2 coefficient is $a-b$. Both must be zero, so addition and subtraction give $a = b = 0$. The polynomials belong to $\mathcal{P}_3$ because their degrees are at most three, even though their actual degree is only two.

Conclusion. Yes, the two polynomials are linearly independent in $\mathcal{P}_3$; they do not form a basis of that larger space.

Exercise 4.3.34: A polynomial dependence by inspection

The identity $(1+t) + (1-t) - 2 = 0$ gives $p_1 + p_2 - p_3 = 0$. Thus p_3 is redundant. To check the remaining pair, a zero combination $a(1+t) + b(1-t) = 0$ has constant coefficient $a+b = 0$ and linear coefficient $a-b = 0$, so $a = b = 0$.

Conclusion. $\{1+t, 1-t\}$ is a basis of the specified span, and the displayed relation proves the original three-function set dependent.

Exercise 4.3.35: Construct a four-vector family with a two-vector basis

Set $v_1 = u_1$, $v_2 = u_1 + u_2$, $v_3 = u_2$, and $v_4 = u_1 - u_2$. Every v_j lies in $\text{Span}\{u_1, u_2\}$, and $v_1 = u_1, v_3 = u_2$ already generate that space. Since a subset of an independent set is independent, $\{u_1, u_2\}$ is independent.

Conclusion. These four choices give $\{v_1, v_3\} = \{u_1, u_2\}$ as a basis of their common span. The additional u_3, u_4 need not be used.

Exercise 4.3.36: Bases of two spaces and their sum

Find a basis for H.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ 0 & -1 & 1 \\ -1 & 1 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Find a basis for K.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \\ 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ -2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Find a basis for their sum.

The sum $H + K$ consists exactly of combinations of all six generators. Use the concatenated matrix, numbering its columns $1, \ldots, 6$ in the order $u_1, u_2, u_3, v_1, v_2, v_3$.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 & -2 & 2 & -1 \\ 2 & 2 & 4 & -2 & 3 & 4 \\ 0 & -1 & 1 & -1 & 2 & 6 \\ -1 & 1 & -4 & 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 & -2 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 5, 6. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$
$$v_4 = (-2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.37: Four independent functions by exact evaluations

Assume $c_1 t + c_2 \sin t + c_3 \cos 2t + c_4 \sin t \cos t = 0$ for every real t. Evaluating at the exact inputs $0, \pi, \pi/2, \pi/4$ gives, successively,

$$\begin{aligned} c_3 &= 0, \\ \pi c_1 + c_3 &= 0, \\ (\pi/2)c_1 + c_2 - c_3 &= 0, \\ (\pi/4)c_1 + (\sqrt{2}/2)c_2 + (1/2)c_4 &= 0. \end{aligned}$$

The first equation gives $c_3 = 0$, the second $c_1 = 0$, the third $c_2 = 0$, and the last $c_4 = 0$. All divisions are by nonzero constants. Thus the only identical zero combination is trivial.

Conclusion. The four functions are linearly independent. Exact evaluation points avoid confusing small numerical roundoff with an actual zero.

Exercise 4.3.38: Seven cosine powers are independent

Suppose $c_0 + c_1 \cos t + \cdots + c_6 \cos^6 t = 0$ for every real t. Choose seven exact inputs $t_j = \arccos r_j$, where

$$(r_1, \ldots, r_7) = (-1, -2/3, -1/3, 0, 1/3, 2/3, 1).$$

Each is allowed since $r_j \in [-1, 1]$. Their evaluation equations are $\sum_{k=0}^{6} c_k r_j^k = 0$ for $j = 1, \ldots, 7$. This is the 7×7 system whose jth row is $[1, r_j, r_j^2, \ldots, r_j^6]$. Its RREF is I_7; equivalently, the polynomial $q(r) = \sum_{k=0}^{6} c_k r^k$ has seven distinct roots but degree at most six, so it must be identically zero. All c_k therefore vanish.

Conclusion. $\{1, \cos t, \ldots, \cos^6 t\}$ is an independent family. The seven evaluations are a finite exact certificate, as requested.